

Lab 1 — Hands-On with AI Tools and Vibe Coding in R

Course: PM599: AI as a Tool for Biostatistics

Date: May 26, 2026 · 1:00–2:50 PM · SSB-114

Instructor: Lu Zhang

Before You Begin

Make sure you have all three platforms open in browser tabs:

- **ChatGPT:** chatgpt.com (sign in with your USC account)
- **Claude:** claude.ai (free account)
- **Gemini:** gemini.google.com (sign in with Google account)

And have **RStudio** (or R) open with the starter script:

```
week1/lab/starter_code/vibe_coding_demo.R
```

Lab Overview

Activity	Time	Description
Setup	1:00–1:10	Account checks, privacy settings, personalization
Activity 1	1:10–1:40	Platform comparison
Activity 2	1:40–2:10	Prompt refinement
Break	2:10–2:15	5 min
Activity 3	2:15–2:45	Vibe coding in R (demo-led)
Wrap-up	2:45–2:50	Debrief and HW1 overview

Setup (1:00–1:10 PM)

Privacy Settings

Follow the instructor's demonstration to configure data privacy settings:

ChatGPT:

1. Click your profile icon (top right) → Settings → Data Controls
2. Turn OFF *"Improve the model for everyone"*

Claude.ai:

- Anthropic does not use Claude.ai conversations to train models by default
- Verify by checking Settings → Privacy

Privacy Settings (cont.)

Gemini:

1. Go to myactivity.google.com
2. Find "Gemini Apps Activity" → turn off activity saving

Why this matters: AI services may use your conversations to retrain models. For research involving preliminary data, unpublished ideas, or anything sensitive, always check privacy settings first.

Personalize Your LLM

Tell each platform who you are so it doesn't assume you are a general user.

ChatGPT — Settings → Personalization → "What would you like ChatGPT to know about you?"

Claude — Paste this at the start of a new conversation:

```
I am a graduate student in biostatistics at USC. I am comfortable with statistical concepts (regression, hypothesis testing, probability) and code in R. Please give technical, precise answers using statistical terminology. When showing code, use R unless I specify otherwise.
```

Try it: Ask each platform "What is the difference between a fixed effect and a random effect?" Notice how responses differ after personalization.

Activity 1 — Platform Comparison (1:10–1:40 PM)

Objective: Compare how different LLMs respond to the same biostatistics prompts.

Format: Individual work, group discussion at the end.

Instructions

Run each of the prompts below through **at least two platforms**. Record the responses (copy-paste into a notes document or screenshot). At the end, compare with a neighbor.

Prompt 1 — Explain a concept:

Explain what a p-value means in plain language. Avoid the phrase "probability that the null hypothesis is true."

Things to look for: Is the explanation accurate? Does it avoid common misconceptions (e.g., confusing p-value with effect size)? Is it clear?

Prompt 2 — Interpret a result:

In a logistic regression model, I get an odds ratio of 2.3 with a 95% confidence interval of (1.1, 4.8) and $p = 0.03$ for the predictor "smoking status." The outcome is 5-year cardiovascular disease. Interpret this result for a clinical audience.

Things to look for: Does it correctly interpret the OR? Does it mention uncertainty (the CI is wide)? Does it note that this is an adjusted or unadjusted estimate?

Discussion Questions

After running the prompts, discuss with your neighbor:

1. Which platform gave the most accurate answer for Prompt 1? Did any make an error?
2. Did any platform add a disclaimer or hedge its answer? Was that appropriate?
3. Would you trust these outputs in a research context without verification? Why or why not?

Note: R-code generation is covered in the Activity 3 demo.

Activity 2 — Prompt Refinement (1:40–2:10 PM)

Objective: Improve a weak prompt step by step and observe how responses change.

Format: Pairs or individual work.

Background

A strong prompt typically provides:

- **Role:** Who should the AI act as?
- **Context:** What is the situation?
- **Task:** What exactly do you need?
- **Format:** How should the answer be structured?

Exercise

Start with this weak prompt:

"Explain confounding."

Then improve it through the following steps. Run each version and note what changes.

Version 2 — Add role, audience, and context:

"You are a biostatistics professor. Explain confounding to a first-year public health MPH student who just learned about regression. Use an example from environmental health research."

Version 3 — Add format:

"You are a biostatistics professor. Explain confounding to a first-year public health MPH student who just learned about regression. Use an example from environmental health research. Structure your answer as:

(1) definition in one sentence, (2) a concrete example, (3) how to address it in analysis, (4) a common mistake to avoid."

Record: How did the quality and relevance of the answer change at each step?
Which addition made the biggest difference for you?

Your Turn

Pick any biostatistics topic you are curious about and write a Version 3-style prompt from scratch. Try it on two platforms. Share with the group if time allows.

Activity 3 — Vibe Coding in R, Demo-Led (2:15–2:45 PM)

Objective: See vibe coding in action — generating, verifying, and correcting R code with AI assistance.

Format: Instructor demonstrates live; students follow along and try one prompt.

Instructor demo (~20 min)

The instructor walks through `week1/lab/starter_code/vibe_coding_demo.R`:

- Section 2: Simulates a clinical trial dataset (SBP before/after)
- Section 3: Box plot of the outcome by group
- Section 4: Two-sample t-test
- At one `[AI PROMPT]` marker: prompt an LLM live, paste, run, verify the output

The golden rule of vibe coding: You are responsible for every line.

AI-generated code that runs is not necessarily *correct* code.

Student turn (~10 min)

Open the starter script in RStudio. Pick **one** [AI PROMPT] marker and run it through ChatGPT, Claude, or Gemini. Paste, run, and verify the result.

Watch for:

- Does the AI use variable names that match your data frame?
- Does it assume packages are loaded that you haven't loaded?
- Does the verbal explanation match what the code actually does?

Continue in HW1: The extension exercise (modify the simulation for a binary outcome / three groups / added covariate) is now part of HW1. See [homework/hw1.md](#) for full instructions.

Wrap-Up and HW1 (2:45–2:50 PM)

Quick debrief — show of hands:

- Who found a factual error in an AI response today?
- Who had AI-generated code that didn't run on the first try?
- Who was surprised by how well (or how poorly) the AI performed?

Homework 1: Compare AI tools on a biostatistics task of your choice.

Full instructions at homework/hw1.md.

Due: May 28, 2026 before class. Submit via Brightspace.

Key Takeaways from Today

1. Different LLMs perform differently on the same prompt — always verify.
2. Prompt quality matters enormously; small changes can dramatically improve responses.
3. AI-generated code runs $\neq$ AI-generated code is correct.
4. Privacy settings are a non-negotiable first step before any research use.
5. You — not the AI — are responsible for the accuracy of your analysis.